

HyRO: Hybrid Retrieval-Augmented Generation Framework for LLMs Integrating Vector Encoding and Knowledge Graphs

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Abstract. Current retrieval-augmented generation methods still exhibit significant limitations in handling complex multi-hop reasoning tasks, mainly because they overly rely on flat vector retrieval over unstructured text and thus fail to explicitly model deep logical relationships between entities. To address this issue, we propose the HyRO framework, a hybrid retrieval-augmented generation framework that synergistically integrates vector encoding and knowledge graphs to achieve synergy between the reasoning capabilities of structured knowledge and the rich semantics of unstructured text. HyRO consists of two core methods: The Automated Hierarchical Knowledge Indexing Method (AHKI) for offline construction of a hierarchical knowledge structure and a vector index; and The Hybrid Collaborative Retrieval and Optimization Method (HCRO) for online query rewriting, hybrid retrieval, and reranking-based fusion to construct a more reliable answer context. Experimental results on a multi-step reasoning test set derived from SQuAD 2.0 show that HyRO consistently outperforms existing methods including RAG, HyDE, and GraphRAG across key metrics. On the Qwen-max model, HyRO achieves 0.83 context precision, 0.93 context recall, 0.76 answer relevance, and 0.93 faithfulness.

Keywords: Community Detection · Hybrid Retrieval · Knowledge Graph · Large Language Models · Multi-hop Reasoning · Retrieval-Augmented Generation · Vector Retrieval

1 Introduction

Large Language Models (LLMs) have achieved remarkable progress in the field of natural language processing, with recent models such as Qwen2.5 [1] and DeepSeek-V3 [2] demonstrating strong performance across diverse benchmarks, and DeepSeek-R1 [3] further advancing reasoning capabilities through reinforcement learning. Yet they still suffer from inherent limitations such as delayed knowledge updates and inadequate factual accuracy. These drawbacks severely restrict their reliability and practicality in knowledge-intensive tasks. Retrieval-Augmented Generation (RAG) technology [4] mitigates the aforementioned issues by integrating information from external knowledge bases, emerging as a

core strategy to enhance the knowledge coverage and timeliness of models. However, existing RAG methods perform poorly when handling complex queries that require multi-step logical reasoning. The fundamental reason lies in their over-reliance on flat vector retrieval mechanisms for unstructured text, which makes it difficult to explicitly model and utilize the deep logical relationships between entities.

To address this challenge, the academic community has proposed two representative optimization approaches. The first category focuses on optimizing the query-document matching mechanism. For example, HyDE [5] reconstructs the query semantic space by generating hypothetical documents, effectively narrowing the semantic gap between queries and target documents. Nevertheless, such methods are still confined to the unstructured text processing framework and fail to fundamentally resolve the issue of insufficient complex reasoning capability. The second category shifts to structured knowledge representation, with GraphRAG [6] as a typical representative. This method constructs knowledge graphs and leverages their topological structures to implement relational reasoning. Although this approach exhibits advantages in mining implicit associations between entities, it inevitably loses rich semantic details from the original text during the graph construction process, resulting in limited performance in tasks that require fine-grained textual evidence for support. In summary, current RAG research is confronted with two core challenges:

1. How to integrate the reasoning capability of structured knowledge while preserving the rich semantics of unstructured text;
2. How to construct a framework that can dynamically balance graph-structured reasoning and textual semantic matching based on query characteristics.

To tackle the above challenges, this paper proposes the HyRO framework—a hybrid retrieval-augmented generation framework for large models that integrates vector encoding and knowledge graphs. By synergistically leveraging the reasoning advantages of structured knowledge and the semantic richness of unstructured text, this framework significantly improves the performance of the system in complex relational reasoning tasks. The main contributions of this paper can be summarized as follows:

1. **Framework:** We design an innovative retrieval-augmented generation framework for large models that fuses vector encoding and knowledge graphs. Through the dynamic construction of knowledge graphs and the synergistic integration of graph retrieval and vector retrieval, HyRO effectively overcomes the limitations of traditional RAG methods in multi-step reasoning tasks.
2. **Methods:** (1) The Automated Hierarchical Knowledge Indexing Method, which combines the semantic understanding capability of large language models with community detection and text summarization techniques, realizing the automatic transformation from unstructured text to multi-level structured knowledge. (2) The Hybrid Collaborative Retrieval and Optimization Method, which proposes a collaborative optimization strategy including

adaptive query rewriting, hybrid retrieval and result reranking, achieving the deep integration of structured knowledge and unstructured semantics to provide highly relevant and information-complete contextual support for language models.

3. **Experiments:** Extensive experiments conducted on the SQuAD 2.0 [7] benchmark dataset demonstrate that HyRO outperforms existing methods across all key metrics including context precision, recall, answer relevance and faithfulness, fully proving the superiority and practical value of the hybrid retrieval strategy in complex question-answering scenarios.

2 Related Work

2.1 RAG Based on Unstructured Text

RAG, proposed by Lewis et al. [4], enhances the generation capability of large language models through dynamic information retrieval from external knowledge bases. Traditional RAG directly uses user queries for vector similarity matching to retrieve relevant text fragments from unstructured document collections as the generation context. However, for queries with complex semantics or ambiguous expressions, this approach struggles to capture the user’s intent, causing a deviation between retrieval results and actual requirements.

To optimize matching quality, subsequent research proposed various query reconstruction strategies. For example, HyDE [5] adopts a core generate-then-retrieve paradigm: it generates hypothetical documents based on the original query, then uses their semantic vectors to complete knowledge retrieval. This bridges the semantic gap between queries and target documents, significantly improving the relevance of retrieval results. Beyond query reconstruction, StructRAG [8] proposes an inference-time hybrid information structurization approach, dynamically converting retrieved documents into optimal structured formats (e.g., tables, graphs) to enhance knowledge-intensive reasoning. Nevertheless, base RAG and HyDE share a core limitation: both adopt a flattened processing approach for unstructured text, lacking explicit modeling capabilities for complex association relationships between entities, thus underperforming in complex tasks such as multi-hop reasoning.

2.2 RAG Based on Structured Knowledge

To overcome the limitations of unstructured methods in logical reasoning, researchers are integrating Knowledge Graphs (KG) into the RAG framework. GraphRAG [6] extracts entities and relations from text to create a knowledge graph, using its structure for information retrieval and relational reasoning. Graph traversal and community detection techniques identify implicit associations and related clusters, enabling multi-step reasoning.

Recent advancements include LightRAG [9], which incorporates graph structures into text retrieval with a dual-level paradigm and supports incremental updates. SubgraphRAG [10] uses lightweight MLPs with a triple-scoring mechanism

for efficient subgraph retrieval, improving accuracy and reducing hallucinations. HippoRAG 2 [11] employs a dual-node graph with Personalized PageRank and LLM-based filtering, enhancing multi-hop QA and associative memory tasks. A study [12] compares RAG and GraphRAG, showing complementary strengths in question answering and summarization.

However, structured knowledge-driven RAG faces challenges. Knowledge graph compression often loses contextual details and fine-grained semantics, and structured graphs struggle with precise text citation and fine-grained matching, impacting retrieval stability. Hybrid strategies are being explored, such as HybGRAG [13], which unifies retrieval-augmented generation, and GeAR [14], which combines graph structures with retrieval agents. These efforts highlight the need for integrating structured and unstructured retrieval to address complex queries, as neither approach alone can fully meet diverse needs.

3 Methodology

3.1 The framework of HyRO

The HyRO framework integrates the semantic retrieval capabilities of vector databases with the structured reasoning advantages of knowledge graphs. HyRO achieves synergy between the reasoning capabilities of structured knowledge and the rich semantics of unstructured text. As shown in Figure 1, the overall framework consists of two methods: The Automated Hierarchical Knowledge Indexing Method (AHKI) and The Hybrid Collaborative Retrieval and Optimization Method (HCRO). AHKI targets the offline indexing stage, aiming to mitigate semantic fragmentation and entity ambiguity in unstructured documents, and to build a hierarchical knowledge structure that supports multi-granular access. HCRO targets the online query stage, aiming to address unstable query expressions and the difficulty of unifying heterogeneous retrieval results, and produces

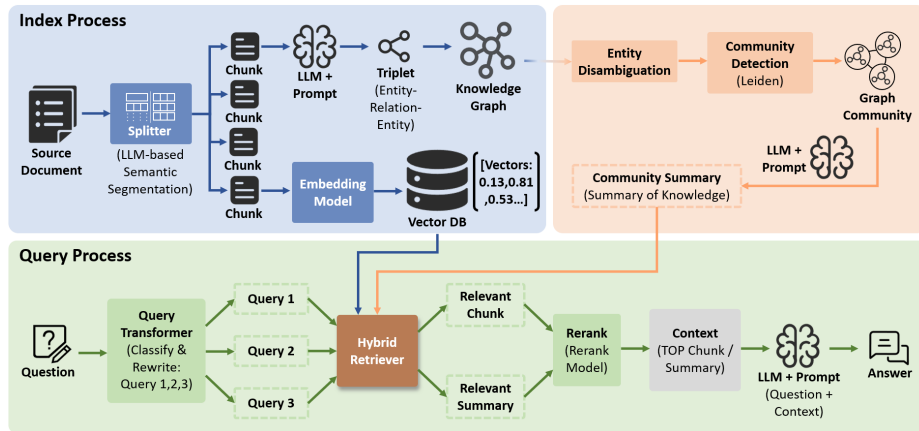


Fig. 1: Overall architecture of the HyRO framework

more reliable answer contexts through hybrid retrieval and optimization-based reranking.

3.2 Automated Hierarchical Knowledge Indexing

This method aims to address key challenges in offline knowledge modeling. First, conventional fixed-window chunking often causes semantic fragmentation, splitting critical information across chunks and reducing recall accuracy. Second, knowledge graphs extracted directly from text frequently suffer from entity ambiguity and relation noise, leading to unstable structured reasoning. In addition, single-granularity indexing cannot simultaneously support fine-grained evidence localization and high-level semantic abstraction. To this end, The Automated Hierarchical Knowledge Indexing Method constructs two types of indices for parallel access, and further builds hierarchical community summaries on the knowledge-graph side to support multi-granular retrieval and reasoning.

Knowledge Graph Construction To alleviate semantic fragmentation, the system first performs semantic-aware document chunking, ensuring that each text chunk corresponds to a coherent semantic unit as much as possible, thereby reducing the probability that key information is broken across chunks. Specifically, an adaptive chunking algorithm based on semantic coherence is adopted: the BGE-large-en-v1.5 embedding model [15] is used to compute cosine similarity between adjacent sentences, and chunk boundaries are identified where the similarity drops by more than 90%. This strategy enhances intra-chunk semantic integrity while meeting the efficiency requirements of subsequent processing.

After obtaining the text chunks, the system uses large language models (LLMs) to extract knowledge triples and construct an initial knowledge graph, explicitly modeling entities and their relations to support subsequent structured retrieval and reasoning. Through structured prompt templates, knowledge triples are extracted from each text chunk, with relationship descriptions required to be temporally invariant to reduce noise introduced by tense variations and context drift. Formally, the knowledge extraction function is defined as:

$$K(c_j) = \{(h_k, r_k, t_k, \text{desc}(h_k), \text{desc}(t_k)) \mid k = 1, 2, \dots, p\} \quad (1)$$

where c_j denotes a text chunk, h_k and t_k denote the head and tail entities, respectively, r_k denotes the relation type, $\text{desc}(\cdot)$ is the entity description function, and p is the number of triples. Aggregating all extracted triples yields the initial graph structure $G_0 = (V_0, E_0)$, where the node set V_0 is the set of unique entities and the edge set E_0 is the set of relation instances.

Knowledge Graph Optimization The initial knowledge graph may still contain entity ambiguity (e.g., same surface form referring to different entities, or different surface forms referring to the same entity) and structural noise introduced by extraction errors, which in turn affects the reliability of graph retrieval

and reasoning. To improve graph quality and support multi-granular access, the system performs two types of optimization: entity disambiguation to increase node consistency, and hierarchical community detection with summary generation to construct retrievable hierarchical semantic units.

(1) Entity disambiguation. The system detects similar entities in the embedding space to form candidate clusters, and combines LLM-based identity verification to merge duplicate entities while reducing the risk of incorrect merges. Specifically, the BGE-M3 model [15] generates entity semantic vectors $\mathbf{v}_{e_i}$, and cosine similarity is computed for entity pairs; when the similarity exceeds the threshold $\theta = 0.95$, temporary links are created to form a similar-entity graph $G_{\text{sim}} = (V_0, E_{\text{sim}})$. The Weakly Connected Components (WCC) algorithm is then applied to G_{sim} to obtain a set of potential duplicate entity clusters $\mathcal{C} = \{C_1, C_2, \dots, C_m\}$. For each cluster C_k , an LLM is used to determine whether the entities refer to the same real-world object, with the decision function defined as:

$$\text{Merge}(C_k) = \begin{cases} \text{True}, & \text{if } \exists \text{ real-world entity } e^* \text{ such that } \forall e_i \in C_k, e_i \text{ refers to } e^* \\ \text{False}, & \text{otherwise} \end{cases} \quad (2)$$

When $\text{Merge}(C_k) = \text{True}$, entities within the cluster are merged, the most complete description is retained, and associated relations are reconnected, yielding the disambiguated knowledge graph.

(2) Hierarchical community detection and summary generation. To support both fine-grained evidence localization and high-level semantic abstraction, the system constructs a hierarchical community structure on the graph and generates a summary representation for each community that can directly participate in retrieval. Considering the long-tail distribution of relation frequencies, the system first applies logarithmic transformation to edge weights to suppress the dominance of extremely frequent relations in community partitioning:

$$w(e) = \log(\text{count}(e) + 1) \quad (3)$$

where $w(e)$ is the weight of edge e , and $\text{count}(e)$ is the relation frequency. The Leiden algorithm [16] is then applied to the disambiguated knowledge graph to optimize community partitioning by maximizing the modularity function Q :

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (4)$$

where $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is the total edge weight, A_{ij} is the adjacency matrix element, $k_i = \sum_j A_{ij}$ is the weighted degree of node i , and $\delta(c_i, c_j)$ is the community indicator function. The algorithm outputs a community hierarchy tree $\mathcal{T} = \{C_0, C_1, \dots, C_L\}$, where C_l denotes the community set at level l , and L is the maximum hierarchy depth. For each community $c \in C_l$, the system uses an LLM to generate a natural-language summary s_c , and enforces an information-density constraint to ensure that the summary can serve as a high-level semantic

representation for retrieval and context construction:

$$\text{InfoDensity}(s_c) \geq \alpha \cdot \text{InfoDensity}\left(\bigcup_{e \in c} \text{desc}(e)\right) \quad (5)$$

where $\text{InfoDensity}(\cdot)$ is the information density metric function, $\alpha = 0.85$ is the threshold, and $\text{desc}(e)$ is the entity description text. With this design, the system can dynamically select the appropriate community granularity according to query requirements, balancing retrieval efficiency, coverage, and semantic precision.

3.3 Hybrid Collaborative Retrieval and Optimization

This method aims to address key challenges in online retrieval and answering. User queries often involve under-specified expressions, ambiguity, and multi-hop reasoning requirements, and single-path retrieval may lead to biased results (e.g., retrieving only similar fragments without structured aggregation, or retrieving only structural summaries without original textual evidence). Meanwhile, candidate results from different knowledge sources are difficult to compare on a unified scale, leading to unstable context fusion. To this end, The Hybrid Collaborative Retrieval and Optimization Method combines query rewriting, dual-path parallel retrieval, and unified reranking to enable collaborative recall of structured and unstructured evidence and to optimize the final context.

Hybrid Collaborative Retrieval The system first enhances query expression through an adaptive query rewriting mechanism, making retrieval closer to user intent and better suited to different reasoning needs. HyRO classifies input queries into four types and applies corresponding strategies to generate rewritten query sets: SIMPLE directly uses the original query; COMPLEX decomposes the problem to support multi-step retrieval; AMBIGUOUS generates multi-perspective rewrites to cover potential intents; CONCEPTUAL adopts the Step-Back strategy to retrieve higher-level conceptual information. Based on the rewritten queries, the system performs structured and unstructured retrieval in parallel. The graph retrieval module targets the hierarchical knowledge graph: it identifies key entity sets in queries via structured output chains, and generates fuzzy matching queries for each entity to accommodate spelling errors; these queries are executed on Neo4j full-text indexes [17], returning relevant community summary sets that provide high-level semantic evidence for aggregation and reasoning. The vector retrieval module preserves fine-grained semantic details of the original text: the BGE-M3 embedding model computes semantic similarity between queries and document chunks, and approximate nearest neighbor search is performed via the Milvus vector database [18], returning relevant text fragments:

$$f_v(q) = \text{TopK}(\text{sim}_{\text{BGE-M3}}(q, c), K = 5)_{c \in \mathcal{C}} \quad (6)$$

where q is the input query, $c \in \mathcal{C}$ denotes a document chunk in the chunk collection, $\text{sim}_{\text{BGE-M3}}(\cdot, \cdot)$ is the semantic similarity function, and $K = 5$ indicates returning the top-5 document fragments most relevant to the query. The two retrieval results are merged to form the raw candidate set, covering both original textual evidence and structured summaries.

Retrieval Optimization To address the difficulty of comparing heterogeneous candidates and the tendency of contexts to become redundant, the system applies a unified reranking module to perform fine-grained relevance modeling and fusion over the candidate set. Specifically, the system employs the BGE-reranker-large cross-encoder [19] to compute relevance scores between the query and candidate documents (either text chunks or community summaries):

$$s(q, d) = \text{softmax} \left(\frac{f_{\text{cross}}(q, d)}{\sqrt{d_k}} \right) \quad (7)$$

where $f_{\text{cross}}(q, d)$ is the raw matching score produced by the cross-encoder for the query-document pair (q, d) , $d_k = 64$ is the dimension of each attention head in the self-attention mechanism, and $\sqrt{d_k}$ scales dot-products to stabilize scoring; the softmax function normalizes scores into a probability distribution on a unified scale, enabling direct comparison and fusion of graph-based and vector-based candidates. The system then selects the top-5 candidates with the highest normalized scores to form the optimized context, improving precise recall of key information while reducing noise.

When a query is decomposed into multiple sub-queries (e.g., COMPLEX or AMBIGUOUS types), the system independently executes the complete hybrid retrieval and reranking procedure for each sub-query, producing corresponding optimized contexts and sub-answers. For single sub-query cases, the sub-answer is directly used as the final output; for multi-sub-query cases, the system guides the large language model with a specially designed aggregation prompt template to synthesize all sub-question-sub-answer pairs into a logically coherent and information-complete final answer. This mechanism allows the system to adapt to query complexity by adjusting retrieval granularity and evidence organization, improving reasoning consistency and factual reliability while maintaining semantic precision.

4 Experiments

4.1 Experimental Setup

Dataset. To comprehensively evaluate system performance on complex multi-step reasoning tasks, our paper adopt the SQuAD 2.0 dataset as the benchmark testing platform. This dataset contains numerous human-annotated Wikipedia articles and their corresponding question-answer pairs, featuring high semantic complexity and factual density. Ours paper selected 512 high-quality Wikipedia

articles to construct the knowledge base, and from over 100,000 original question-answer pairs, we filtered 10,000 questions requiring multi-step reasoning as the test set through expert annotation. The filtering criteria were strictly defined as: (1) each question must involve at least two logical steps; (2) answering requires integrating information scattered across different paragraphs; and (3) the answer cannot be directly extracted from a single text fragment. This rigorous filtering process ensures that the evaluation focuses on the system’s complex reasoning capabilities rather than simple retrieval performance.

Baselines. To comprehensively validate the effectiveness of HyRO, we compare it with three representative RAG methods: (1) RAG: a standard retrieval-augmented generation framework; (2) HyDE: a hypothetical document embedding method that reconstructs the query semantic space by generating hypothetical answers; and (3) GraphRAG: a knowledge graph-based retrieval framework that leverages graph structures for relational reasoning.

Evaluation Metrics. Evaluation employs four key metrics provided by the RAGAS framework [20]: (1) Context Precision(CP): measures the semantic relevance between retrieved content and the question; (2) Context Recall(CR): evaluates the completeness of retrieved results in covering information required for answering; (3) Answer Relevance(AR): quantifies the semantic matching degree between generated answers and questions; and (4) Faithfulness(F): measures whether answers strictly rely on retrieved content rather than the model’s internal knowledge.

Implementations. Vector indexing was implemented using Milvus 2.2.11 [18], and the knowledge graph was stored in the Neo4j 5.24.0 graph database [17]. For embedding models, BGE-M3 was used for document encoding, and BGE-reranker-large was employed for result reranking. Regarding large language models, our paper selected Qwen-max and DeepSeek-V3 as base models. Key hyperparameter settings are as follows: maximum text chunk length of 512 tokens, top- k for retrieval results set to 5, top- p set to 0.9, temperature set to 0, and the similarity threshold for the Leiden community detection algorithm uniformly set to 0.95.

4.2 Performance comparison

Our paper conducted a comprehensive evaluation of HyRO against representative baselines on the SQuAD 2.0 dataset using two large language models (Qwen-max and DeepSeek-V3). As shown in Table 1, HyRO achieves consistently strong overall performance across both backbones. On Qwen-max, HyRO obtains the highest CP (0.83) and AR (0.76) while maintaining competitive CR (0.93), indicating that it can provide more accurate and relevant context without sacrificing coverage. On DeepSeek-V3, HyRO achieves the best results on all

metrics (CP=0.80, CR=0.92, AR=0.67, F=0.92), demonstrating that the hybrid retrieval design generalizes well across different LLMs and remains effective under more challenging settings.

Table 1: Performance comparison on the SQuAD 2.0 dataset

LLM	Method	CP	CR	AR	F
Qwen-max	RAG	0.81	0.93	0.66	0.87
	HyDE	0.74	0.88	0.72	0.90
	GraphRAG	0.51	0.92	0.72	0.95
	HyRO (ours)	0.83	0.93	0.76	0.93
DeepSeek-V3	RAG	0.58	0.59	0.61	0.80
	HyDE	0.75	0.90	0.65	0.89
	GraphRAG	0.61	0.72	0.62	0.91
	HyRO (ours)	0.80	0.92	0.67	0.92

4.3 Ablation Study

To quantify the contribution of each component in HyRO, our paper performed incremental ablation experiments on SQuAD 2.0 with Qwen-max by progressively adding The Automated Hierarchical Knowledge Indexing Method (AHKI), Hybrid Collaborative Retrieval (HCR), and Retrieval Optimization (RO) on top of the RAG baseline. Table 2 shows that introducing AHKI substantially improves AR (0.66→0.73), suggesting that structured knowledge benefits answer relevance, but it also leads to sharp drops in CP and CR (to 0.11 and 0.39), indicating that relying on graph summaries alone is insufficient for high-quality retrieval. Adding HCR greatly improves coverage and reliability (CR=0.96, F=0.92). On top of that, RO further refines the retrieved context by removing noise and reranking candidates, yielding the best overall trade-off: CP increases to 0.83 while CR remains at 0.93, together with the highest AR (0.76) and F (0.93).

Table 2: Ablation study of different components

Method	CP	CR	AR	F
RAG	0.81	0.93	0.66	0.87
RAG(+AHKI)	0.11	0.39	0.73	0.82
RAG(+AHKI+HCR)	0.32	0.96	0.74	0.92
RAG(+AHKI+HCR+RO) (Full framework)	0.83	0.93	0.76	0.93

4.4 Parameter Analysis

Table 3: Hyperparameter sensitivity on SQuAD 2.0 with Qwen-max. The default settings used in the main experiments are $\text{top-}k = 5$, $\theta = 0.95$, and $B = 90$ (where B denotes the chunking Breakpoint).

(a) $\text{top-}k$					(b) θ					(c) B				
k	CP	CR	AR	F	θ	CP	CR	AR	F	B	CP	CR	AR	F
3	0.79	0.88	0.73	0.91	0.85	0.78	0.90	0.72	0.90	70	0.75	0.89	0.70	0.88
5	0.83	0.93	0.76	0.93	0.90	0.81	0.92	0.74	0.92	80	0.80	0.91	0.73	0.91
7	0.80	0.94	0.75	0.92	0.95	0.83	0.93	0.76	0.93	90	0.83	0.93	0.76	0.93
10	0.76	0.95	0.73	0.90	0.99	0.80	0.91	0.74	0.92	95	0.81	0.92	0.74	0.93

To examine the robustness of HyRO with respect to key hyperparameters, our paper conducted sensitivity experiments on SQuAD 2.0 using Qwen-max. In each group, only one parameter was varied while the other settings were fixed to the default configuration used in the main experiments ($\text{top-}k = 5$, similarity threshold $\theta = 0.95$, and $B = 90$, where B denotes the chunking Breakpoint). Table 3 reports the results in terms of Context Precision (CP), Context Recall (CR), Answer Relevance (AR), and Faithfulness (F), showing that the default setting achieves the best or near-best overall trade-off: $\text{top-}k = 5$ yields the highest CP and AR with strong CR and F, $\theta = 0.95$ provides the most balanced thresholding effect, and the chunking parameter B performs best at $B = 90$ compared to smaller values that reduce contextual completeness and larger values that introduce redundancy. Therefore, our paper adopt $\text{top-}k = 5$, $\theta = 0.95$, and $B = 90$ as the default configuration in the full experiments.

5 Conclusion

This paper presents HyRO, a large language model retrieval-augmented generation framework that integrates vector retrieval and knowledge graphs, designed to address the limitations of traditional RAG methods when handling complex multi-hop reasoning tasks. Through the Automated Hierarchical Knowledge Indexing Method and the Hybrid Collaborative Retrieval and Optimization Method, HyRO achieves synergy between the reasoning capabilities of structured knowledge and the rich semantics of unstructured text. Experimental results on the SQuAD 2.0 dataset demonstrate that HyRO outperforms baseline methods including RAG, HyDE, and GraphRAG across key metrics such as context precision, answer relevance, and faithfulness. Ablation studies further validate the effectiveness of each framework component, demonstrating HyRO’s ability to balance retrieval comprehensiveness, result precision, and system scalability.

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